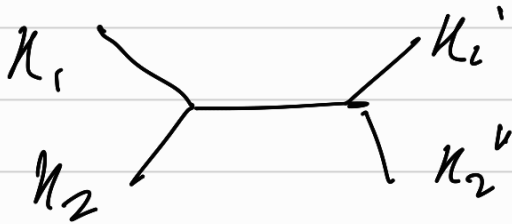


$\phi\phi \rightarrow \phi\phi$ in ϕ^3 :

$$\tilde{\mathcal{L}} = g^2 \left(\frac{1}{(k_1 + k_2)^2 + m^2} + \frac{1}{(k_1 - k_1')^2 + m^2} + \frac{1}{(k_1 - k_2')^2 + m^2} \right)$$



$$k_1 + k_2 = k_1' + k_2'$$

all on-shell $k_i^2 = -m_i^2$

C.O.M frame we take $\vec{k}_1 + \vec{k}_2 = 0$

i.e. $\vec{k}_1 = -\vec{k}_2$

$$\vec{k}_1 = (0, 0, |k_1|)$$

total En: $E_1 + E_2$

$$\vec{k}_2 = (0, 0, -|k_1|)$$

$$S \equiv -(k_1 + k_2)^2$$

$$\bullet (k_1 + k_2)^M = (E_1 + E_2, 0, 0, 0)^M$$

$$\therefore S = (E_1 + E_2)^2 \quad \leftarrow \text{C.O.M Energy Squares}$$

$$E_1 = (|\vec{k}_1|^2 + m_1^2)^{1/2}$$

$$E_2 = (|\vec{k}_1|^2 + m_2^2)^{1/2}$$

• Thus, we can solve for $|k_1|$ in terms of S .

$$|\vec{k}_1| = \frac{1}{2\sqrt{S}} \left(S^2 - 2(m_1^2 + m_2^2)S + (m_1^2 - m_2^2)^2 \right)^{1/2}$$

(CM-frame)

outcome: $\vec{k}_1 + \vec{k}_2 = 0$ and $(E_1 + E_2)^2 = s$

$$|\vec{k}_1| = \frac{1}{2\sqrt{s}} \left(s^2 - 2(m_1^2 + m_2^2)s + (m_1^2 - m_2^2)^2 \right)^{1/2}$$

only need θ between k_1 and k_1'

$$t = -(\vec{k}_1 - \vec{k}_1')^2 = ((k_1)_0 - (k_1')_0)^2 - |\vec{k}_1 - \vec{k}_1'|^2$$

$$(E_1 - E_1')^2 = \vec{k}_1 \cdot \vec{k}_1 - \vec{k}_1 \cdot \vec{k}_1' + 2\vec{k}_1 \cdot \vec{k}_1'$$

$$= (E_1 - E_1')^2 = |\vec{k}_1|^2 - |\vec{k}_1'|^2 + 2\vec{k}_1 \cdot \vec{k}_1'$$

$$= E_1^2 + E_1'^2 - 2E_1 E_1' - |\vec{k}_1|^2 - |\vec{k}_1'|^2 + 2\vec{k}_1 \cdot \vec{k}_1'$$

since

$$E_1^2 - |\vec{k}_1|^2 = m_1^2$$

$$t = m_1^2 + m_2^2 - 2E_1 E_1' + 2|\vec{k}_1| |\vec{k}_1'| \cos(\theta)$$

$$\mathcal{L} = g^2 \left(\frac{2}{m^2 - s} + \frac{1}{m^2 - t} + \frac{1}{m^2 - u} \right)$$

We can consider the Lab-frame:

• Particle 2 is at rest $\vec{k}_2 = \vec{0}$

then

$$s = -(\vec{k}_1 + \vec{k}_2)^2, \quad E_1^2 = k_1^2 + m_1^2, \quad E_2 = m_2$$

$$\vec{k}_2 = (m_2, 0, 0, 0)$$

$$s = -(\vec{k}_1 + \vec{k}_2)_0^2 - |\vec{k}_1|^2$$

$$= (E_1 + E_2)^2 - |\vec{k}_1|^2$$

$$= E_1^2 + 2E_1 E_2 + E_2^2 - |\vec{k}_1|^2$$

$$= |\vec{k}_1|^2 + m_1^2 + 2(|\vec{k}_1|^2 + m_1^2)^{1/2} m_2 + m_2^2 - |\vec{k}_1|^2$$

$$s = (m_1^2 + m_2^2) + 2(|\vec{k}_1|^2 + m_1^2)^{1/2} m_2$$

$$\frac{(S - (m_1 + m_2))^2}{(2m_2)^2} - m_1^2 = |\vec{k}_1|^2$$

$$\left(S^2 - 2(m_1^2 + m_2^2)S + (m_1^2 + m_2^2)^2 - 4m_1^2 m_2^2 \right)^{1/2}$$

$$\begin{aligned} \text{with } & (m_1^4 + 2m_1^2 m_2^2 + m_2^4 - 4m_1^2 m_2^2) \\ & = (m_1^4 - 2m_1^2 m_2^2 + m_2^4) \\ & = (m_1^2 - m_2^2)^2 \end{aligned}$$

$$\therefore |k_1| = \frac{1}{2m_2} \sqrt{S^2 - 2(m_1^2 + m_2^2)S + (m_1^2 - m_2^2)^2} \quad (\text{Lab frame})$$

$$\text{With } m_2 |k_2|_{\text{Lab}} = \sqrt{S'} |k|_{\text{cm}}$$

* Experiment is done in a volume V for some time T .

$$\langle f|i \rangle = (2\pi)^4 \delta^4(k_{\text{in}} - k_{\text{out}}) i \mathcal{Z}$$

* To get a probability $\langle f|i \rangle$: square and divide by the norms of i and f :

$$P = \frac{|\langle f|i \rangle|^2}{\langle f|f \rangle \langle i|i \rangle}$$

$$|\langle f|i \rangle|^2 = [(2\pi)^4 \delta^4(k_{\text{in}} - k_{\text{out}})]^2 |\mathcal{Z}|^2$$

$$\hookrightarrow (2\pi)^4 \delta^4(k_{\text{in}} - k_{\text{out}}) \times (2\pi)^4 \delta^4(0)$$

$$\text{with } (2\pi)^4 \delta(0) = \int d^4x e^{i0x} = VT \leftarrow \int d^3x \int dt$$

$$\langle k|k \rangle = (2\pi)^3 (2k^0) \delta^3(\vec{0}) = 2k^0 V$$

$$\langle i|i \rangle = 4E_1 E_2 V^2$$

$$\langle f|f \rangle = \prod_{j=1}^{n'} 2(k_j^0) V$$

Thus,

$$P = \frac{(2\pi)^4 \delta^4(k_{\text{in}} - k_{\text{out}}) VT |\mathcal{Z}|^2}{(4E_1 E_2 V^2) \prod_{j=1}^{n'} 2k_j^0 V}$$

$$(4 E_1 E_2 V^{-1}) \left(\prod_{j=1}^{n'} 2 \pi \delta^3(\vec{k}_j - \vec{k}_j') \right) V$$

$$\boxed{\dot{P} = \frac{P}{T}}$$

• To get something measurable: sum each outgoing $\vec{k}_j$ over some small range.

$$\vec{k}_j' = \left(\frac{2\pi}{L}\right) \vec{n}_j' \quad V = L^3 \quad \vec{n}_j' \in \mathbb{Z}^3$$

in the limit $L \rightarrow \infty$

$$\sum_{\vec{n}_j'} \rightarrow \frac{V}{(2\pi)^3} \int d^3 k_j'$$

$$\therefore \dot{P} = \frac{(2\pi)^4 \delta^4(k_{in} - k_{out}) |\mathcal{M}|^2}{4 E_1 E_2 V} \prod_{j=1}^{n'} \frac{1}{(2\pi)^3} d^3 k_j'$$

where

$$d^3 k_j' = \frac{d^3 u}{(2\pi)^3 2k^0} \quad \left(\text{Lorentz inv. phase space differential} \right)$$

• To get $\dot{P}$: prob. per unit time to scatter into n' particles

to be a cross - sec. we must divide by the incident flux

$$\Phi = \frac{(\# \text{ of particles})}{(\text{Volume}) \cdot |V_{in}|}$$

"Velocity flux"

• Here: we have 1 incident particle (in)

• Volume: V

• Velocity: $v = |\vec{k}_1| / E_1$

$$\Phi = \frac{|\vec{k}_1|}{E_1 V}$$

$$\prod_{j=1}^{n'} \frac{1}{(2\pi)^3} d^3 k_j'$$

$$\frac{P}{\Phi} = \frac{(2\pi)^4 \delta^4(k_{in} - k_{out}) |\mathcal{E}|^2}{4 |k_1|_{\text{lab}} m_2} \prod_{j=1}^n d^3k_j$$

but since $|k_1|_{\text{lab}} m_2 = |k|_{\text{cm}} \sqrt{s}$

$$d\sigma = \frac{1}{4 |k_1|_{\text{cm}} \sqrt{s}} |\mathcal{E}|^2 d\text{LIPS}_{n'}(k_1 + k_2)$$

$$d\text{LIPS}_{n'}(k) \equiv (2\pi)^4 \delta^4(k - \sum_{i=1}^{n'} k_i') \prod_{j=1}^{n'} d^3\tilde{k}_j'$$

$\Rightarrow$ $2 \rightarrow n'$ Scattering

* $2 \rightarrow 2$ Scattering

$$d\text{LIPS}_2(k) = (2\pi)^4 \delta^4(k - k_1' - k_2') d^3\tilde{k}_1' d^3\tilde{k}_2'$$

$k \equiv k_1 + k_2$

• Can compute dLIPS in any frame as it is Lorentz. invariant.

CM frame: $\vec{k} = \vec{k}_1 + \vec{k}_2 = \vec{0}$
 $k^0 = E_1 + E_2 = \sqrt{s}$

$$d\text{LIPS}_2(k) = \frac{1}{4(2\pi)^2 E_1' E_2'} \delta(E_1' + E_2' - \sqrt{s}) \delta^3(k_1' + k_2') d^3k_1' d^3k_2'$$

Use $\delta^3(k_1' + k_2')$ to integrate over d^3k_2'

$$d\text{LIPS}_2(k) = \frac{1}{4(2\pi)^2 E_1' E_2'} \delta(E_1' + E_2' - \sqrt{s}) d^3k_1'$$

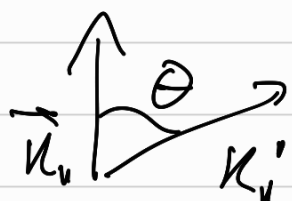
$$E_1' = \sqrt{k_1'^2 + m_1^2} \quad \text{etc.}$$

$$d^3k_1' = |k_1'|^2 d|k_1'| d\Omega_{\text{cm}}$$

where

$d\Omega_{cm} = \sin(\theta) d\theta d\phi$: differential solid

θ : angle between $\vec{k}_2$ and $\vec{k}_1'$ in CM frame



$$E_2' = |\vec{k}_1'|^2$$

$$\vec{k}_1' = -\vec{k}_2'$$

• Basically writing $\vec{k}_1'$ in spherical coordinates taking $\vec{k}_2$ as the "x-axis"

$$d\text{LIPS}_2(k) = \left(\frac{1}{4(2\pi)^2 E_1' E_2'} \right) \delta(E_1' + E_2' - \sqrt{s}) |\vec{k}_1'|^2 d\vec{k}_1' d\Omega$$

$$S = S(k_u)$$

$$\int d\text{LIPS}_2(k) = \frac{1}{4(2\pi)^2 E_1' E_2'} \left(\int \delta(E_1' + E_2' - \sqrt{s}) |\vec{k}_1'|^2 d\vec{k}_1' \right) d\Omega$$

Since $\int_{-\infty}^{\infty} dx f(x) \delta(g(x)) = \sum_i \frac{f(x_i)}{|g'(x_i)|}$

$$g(|\vec{k}_1'|) = E_1' + E_2' - \sqrt{s}$$

$$\therefore \frac{\partial}{\partial |\vec{k}_1'|} (E_1' + E_2' - \sqrt{s}) = \frac{|\vec{k}_1'|}{E_1'} + \frac{|\vec{k}_1'|}{E_2'}$$

$$= |\vec{k}_1'| \frac{(E_1' + E_2')}{E_1' E_2'} = \frac{\sqrt{s} |\vec{k}_1'|}{E_1' E_2'}$$

$$\therefore d\text{LIPS}_2(k) = \left(\frac{E_1' E_2'}{\sqrt{s} |\vec{k}_1'|} \right) \left(\frac{1}{4(2\pi)^2 E_1' E_2'} \right) |\vec{k}_1'|^2 d\Omega_{cm}$$

$$= \frac{|\vec{k}_1'|}{\sqrt{s} 16\pi^2} d\Omega_{cm}$$

$$\therefore \boxed{\frac{d\sigma}{d\Omega_{cm}} = \frac{1}{64\pi^2 s} \frac{|K_i'|}{|K_i|} |\mathcal{T}|^2}$$

$$\bullet dt = 2|K_i| |K_i'| d\cos\theta = 2|K_i| |K_i'| \frac{d\Omega_{cm}}{2\pi}$$

$$\frac{d\sigma}{dt} = \frac{1}{64\pi s |K_i|^2} |\mathcal{T}|^2$$

* For $2 \rightarrow n'$ can define a total cross-section by int. completely over all outgoing momenta and dividing by symm factor S .

$$S = \prod_i n_i! \quad (\text{identical particles})$$

$$\sigma = \frac{1}{S} \int d\sigma$$

• For $2 \rightarrow 2$,

$$\sigma = \frac{1}{S} \int d\Omega_{cm} \frac{d\sigma}{d\Omega_{cm}} = \frac{2\pi}{S} \int_{-1}^1 d\cos\theta \frac{d\sigma}{d\Omega_{cm}}$$

$$S = 2$$

Scattering in d^3 : $\sqrt{s} = E_1 + E_2$
 mb since all masses are equal
 $E_i = E$

$$E = \frac{1}{2}\sqrt{s}$$

$$\text{and } |K_i| = |K_i'| = \frac{1}{2}(s - 4m^2)^{1/2}$$

$$\begin{aligned} \sigma &= 2m^2 = 2\left(\frac{1}{2}\sqrt{s}\right)\left(\frac{1}{2}\sqrt{s}\right) + 2\frac{1}{4}(s - 4m^2)\cos\theta \\ &= 2m^2 - \frac{1}{2}s + \frac{1}{2}(s - 4m^2)\cos\theta \\ &= -\frac{1}{2}(s - 4m^2) + \frac{1}{2}(s - 4m^2)\cos\theta \\ &= \frac{1}{2}(s - 4m^2)(\cos\theta - 1) \\ &= -\frac{1}{2}(s - 4m^2)(1 - \cos\theta) \end{aligned}$$

$$v_1 = -\frac{1}{2} (s - 4m^2) (1 + \cos(\theta))$$

$$\therefore \tau = \frac{E_g^2}{3u^2} \left[1 - \frac{8}{15} \left(\frac{s - 4m^2}{m^2} \right) + \frac{8}{15} (\dots) \right]$$

Ex 1] Consider a theory w two real scalar fields A and B with an interaction term

$$\mathcal{L}_I = g A B^2 \quad | \quad m_A > 2m_B$$

$$(a) \quad d\Gamma = \frac{1}{2E_1} |\mathcal{M}|^2 d\text{LIPS}(k_1)$$

$$s = -k_1^2 = m_1^2$$

$$E_1 = m_1$$

$$\Gamma = \frac{1}{s} \int \frac{1}{2E_1} |\mathcal{M}|^2 d\text{LIPS}(k_1)$$

$$d\text{LIPS}(k_1) = \frac{|k_1|}{16\pi^2 \sqrt{s}} d\Omega_{cm}$$

$$\Gamma = \frac{1}{s} \int \frac{1}{2E_1} |\mathcal{M}|^2 \left(\frac{|k_1|}{16\pi^2 \sqrt{s}} \right) d\Omega_{cm}$$

$$|k_1| = \frac{1}{2\sqrt{s}} \left(s^2 - 2(m_1^2 + m_2^2)s + (m_1^2 - m_2^2)^2 \right)^{1/2}$$

$$\begin{aligned} \Gamma &= \frac{1}{s} \int \frac{1}{2E_1} |\mathcal{M}|^2 \left(\frac{1}{16\pi^2 m_1} \right) \left(\frac{1}{2m_1} \right) \left(m_1^4 - 2(m_1^2 + m_2^2)m_1^2 + (m_1^2 - m_2^2)^2 \right)^{1/2} d\Omega_{cm} \\ &= \left(\frac{1}{s} \right) \left(\frac{1}{64\pi^2 m_1^3} \right) \int d\Omega_{cm} \left(m_1^4 - 2m_1^2 \dots \right)^{1/2} |\mathcal{M}|^2 \end{aligned}$$

A $\rightarrow$ '2B' decay

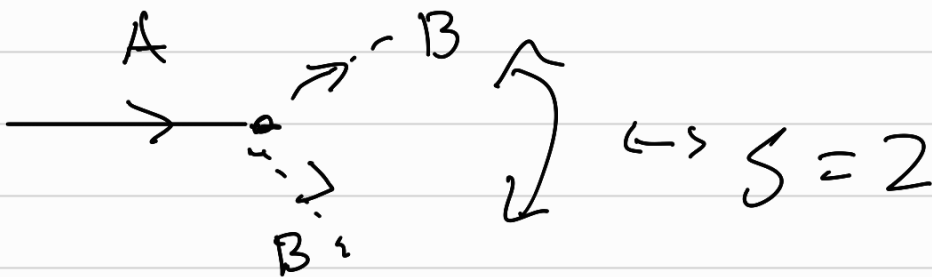
$$m_1 = m_2 = m$$

$$\sqrt{M_A^4 - 4M_B^2 M_A^2} = M_A^2 \left(1 - 4 \frac{M_B^2}{M_A^2}\right)^{1/2}$$

$$\therefore \Gamma = \left(\frac{1}{S}\right) \left(\frac{M_A^2}{64\pi^2 M_A^3}\right) \left(1 - 4 \frac{M_B^2}{M_A^2}\right)^{1/2} \int d\Omega |\tau|^2$$

$$\Gamma = \left(\frac{1}{S}\right) \left(\frac{\left(1 - 4 \frac{M_B^2}{M_A^2}\right)^{1/2}}{64\pi^2 M_A}\right) \int d\Omega |\tau|^2$$

A $\rightarrow$ 2B Decay



Vertex: ig
 $S=2$

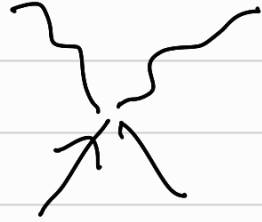
$$\therefore |\tau|^2 = |ig|^2 = 4g^2$$

$$\Gamma = g^2 \frac{\left(1 - 4 \frac{M_B^2}{M_A^2}\right)^{1/2}}{32\pi^2 M_A} \int d\Omega$$

4π

$$\Gamma = \frac{g^2}{8\pi M_A} \left(1 - 4 \frac{M_B^2}{M_A^2}\right)^{1/2}$$

11.2 (Compton Scattering)



$$K_e = (E_e, 0, 0, 0)$$

$$K_\gamma = (E_\gamma, 0, 0, E_\gamma)$$

$$K_e^2 = -m^2$$

$$K_\gamma^2 = -m_\gamma^2 = 0$$

$$S = -(K_\gamma + K_e)^2 = -K_\gamma^2 - 2K_\gamma K_e - K_e^2$$

$$S = m^2 = 2(K_\gamma)_\mu (K_e)_\mu$$

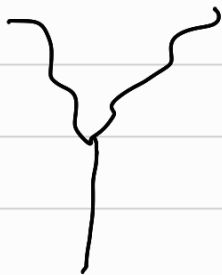
$$= m^2 + 2E_\gamma m$$

Let $E_\gamma = \omega$, then

$$S = m^2 + 2\omega m$$

$$u = -(K_e - K_\gamma')^2$$

$$= m^2 - 2\omega' m$$



$$K_\gamma = (\omega, 0, 0, \omega)$$

$$K_\gamma' = (\omega', \vec{\omega}')$$

$$s + t + u = 2m^2$$

$$t = 2m^2 - m^2 + 2\omega' m - m^2 - 2\omega m$$

$$t = 2\omega' m - 2\omega m$$

$$= 2m(\omega' - \omega)$$

$$= -2\omega\omega' + 2|K_\gamma||K_\gamma'| \cos(\theta_{FT})$$

$$\frac{2m(\omega' - \omega) + 2\omega\omega'}{2|K_\gamma||K_\gamma'|} = \cos(\theta_{FT})$$

Both can be s.t. $|K_\gamma| = \omega$
 $|K_\gamma'| = \omega'$

$$\cos(\theta_{FT}) = \frac{m(\omega' - \omega)}{\omega\omega'} + 1$$

$$\cos(\theta_{FT}) = 1 - m \left(\frac{1}{\omega'} - \frac{1}{\omega} \right)$$

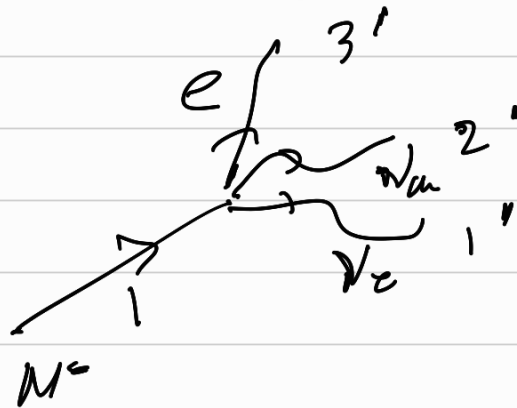
$$\frac{d\sigma}{d\Omega_{CF}} = \frac{d\sigma}{dt} \frac{dt}{d\Omega_{CF}}$$

$$dt = 2\omega'^2 \cos(\theta)$$

11.3 Muon decay: $\mu^- \rightarrow e^- \bar{\nu}_e \nu_\mu$

$$|\mathcal{M}|^2 = 64 G_F^2 (k_1 \cdot k_2') (k_1' \cdot k_3')$$

$$\begin{aligned} k_1 &: \mu^- \\ k_1' &: \bar{\nu}_e \\ k_2 &: \nu_\mu \\ k_3' &: e^- \end{aligned}$$



$$\Gamma = \frac{32 G_F^2}{m} \int (k_1 \cdot k_2') (k_1' \cdot k_3') d\text{LIPS}_3(k_i)$$

$$k_1 = (m, \vec{0})$$

$$\begin{aligned} m_\mu &= 0 \\ m_e &= 0 \end{aligned}$$

$$(a) \quad d\text{LIPS}_3(k_1) = (2\pi)^4 \delta^4(k_1 - (k_1' + k_2' + k_3')) \prod_{j=1}^3 \left(\frac{d^3 k_j}{(2\pi)^3 2k_j^0} \right)$$

$$= (2\pi)^4 \delta^4(k_1 - k_1' - k_2' - k_3') d^3 k_1' d^3 k_2' d^3 k_3'$$

$$= (2\pi)^4 \delta(E_1 - E_1' - E_2' - E_3') \delta^3(\vec{k}_1 - \vec{k}_1' - \vec{k}_2' - \vec{k}_3')$$

while

$$dLIPS_2(k_1 - k_3') = (2\pi)^4 \delta^4(k_1 - k_3' - k_1' - k_2') d\tilde{k}_1' d\tilde{k}_2'$$

and so

$$dLIPS_3(k_1) = d\tilde{k}_3' dLIPS_2(k_1 - k_3')$$

$$\therefore \Gamma = \frac{32G_F^2}{m} \int d\tilde{k}_3' K_{2\mu} K_{3\nu} \int k_2'^\mu k_3'^\nu dLIPS_2(k_1 - k_3')$$

✓

(b) By Lorentz inv.

$$\int k_2'^\mu k_3'^\nu dLIPS_2(k_1 - k_3') = A k^2 g^{\mu\nu} + B k^\mu k^\nu$$

$$\bullet \int dLIPS_2(k) \stackrel{?}{=} \int \frac{|k_1'|}{16\pi^2 \sqrt{s}} d\Omega_{cm}$$

$$\sqrt{s} = m$$

$$|k_1'| = \frac{m'}{2} \quad ?$$

$$k_1 = k_1' + k_2' + k_3'$$

$$-m^2 = k_1 k_2 +$$

$$\frac{1}{32\pi^2} \int d\Omega_{cm} = \frac{1}{8\pi}$$

ir